Evolutionary formulations for design of heterogeneous Earth observing constellations

Pau Garcia Buzzi
Department of Aerospace Engineering
Texas A&M University
College Station, TX 77843
pg428@tamu.edu

Daniel Selva
Department of Aerospace Engineering
Texas A&M University
College Station, TX 77843
dselva@tamu.edu

***Abstract*—The constellation and orbit design problem for Distributed Spacecraft Missions is challenging due to the extremely large space of alternatives, the presence of multiple conflicting objectives, and the expensive evaluation functions required to assess these objectives. Prior work in the literature has typically constrained this problem to simple constellation patterns that generally provide good coverage performance, such as the homogeneous Walker Delta pattern. However, this ignores potentially good architectures formed by heterogeneous constellations mixing satellites at different altitudes and inclinations. Thus, this paper studies methods to explore this region of the constellation design space which is understudied and has potential to be a cost-efficient way of satisfying regional coverage requirements and solving orbit design problems with disjoint areas of interest. We adopt a framework based on multi-objective evolutionary algorithms, which have been used extensively in the literature to solve similar problems due to their ability to deal with mixed-integer problems with highly non-linear and non-convex objective functions such as the numerical simulations required to compute coverage performance for multi-satellite systems. Moreover, we focus on formulations (i.e., chromosomes and operators) rather than algorithmic details, since it is well known that formulations are at least as critical in driving performance as the details of the optimization algorithm used itself. Specifically, we explore two new formulations for heterogeneous constellation design: (1) A fixed-length chromosome containing m 5-tuples, which encode a hybrid architecture with a fixed number of m Walker constellations at different altitudes and inclinations, where each constellation is defined by a 5-tuple consisting of altitude, inclination, number of satellites, number of planes and relative spacing parameter. In this formulation, the number of satellites in each constellation can be set to 0 to effectively reduce the number of constellations. (2) A variable-length chromosome, which encodes a heterogeneous constellation defined by the total number of satellites n and p 2-tuples, where each plane is defined by a 2-tuple consisting of its altitude and inclination and includes n/p satellites. All planes are assumed to be equally spaced in RAAN and satellites within a plane are equally spaced in mean anomaly. These two formulations are compared to a third formulation proposed earlier in the literature, which consists of a variable-length chromosome containing n 4-tuples to encode a constellation of n satellites, where each satellite is defined by a 4-tuple with its altitude, inclination, RAAN, and mean anomaly. The performance of these formulations is compared by solving a constellation design problem where the objectives are to optimize coverage performance and lifecycle cost. Search performance is assessed by looking at the evolution of hypervolume with the number of function evaluations. The formulations discussed in this paper were developed in the context of a NASA-funded project to develop a tool called Tradespace Analysis Tool for Constellations using Machine Learning (TAT-C ML). This tool is intended to be used during early stages of the design of Earth observation missions, and it is planned to be released open source within the next year.**

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1. INTRODUCTION

The constellation and orbit design problem for missions that contain multiple satellites, commonly known as Distributed Spacecraft Missions (DSMs), is challenging due to two main reasons. First, the number of possible architecture designs grows extremely fast with the number of satellites in the constellation. This happens because the number of design parameters increases very fast with the constellation size, leading to combinatorial explosion. Second, evaluating the coverage performance– or other quantities derived from it such as robustness metrics– of just a single design can be very computationally expensive. Therefore, the full factorial enumeration and evaluation of all possible designs becomes infeasible, and more sophisticated sampling techniques for tradespace exploration or optimization are needed to solve this problem.

To overcome these difficulties, a common approach found in the literature is to restrict the design space to certain constellation geometries that generally provide good coverage performance thanks to their symmetry characteristics (e.g., Walker Delta pattern). By doing so, the design space gets considerably reduced, making the problem itself more tractable. However, other constellation geometries that do not follow any of these well-known patterns could potentially outperform those that follow them [1].

Evolutionary algorithms have been proven to be an effective way to solve the orbit design problem [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15] thanks to their ability to deal with mixed-integer problems with highly non-linear and non-convex objective functions, such as the numerical simulations needed to evaluate coverage performance for satellite constellations [16], [17], [18]. Particularly, multi-objective evolutionary algorithms are generally used since, in constellation design, there exist multiple conflicting objectives to optimize and the preferences between them are problem-specific and hard to define a priori. For instance, in addition to the typical trade-off between the cost and performance (in this case, coverage performance) there may also be more subtle trade-offs between different coverage metrics, such as percent coverage vs revisit time, or mean vs maximum revisit time.

This work presents two new evolutionary formulations to explore heterogeneous constellations containing satellites at different altitudes and inclinations, while preserving a certain level of constellation symmetry in other orbital parameters. The main motivation behind these two formulations is to explore a part of the constellation design space that is usually understudied and has potential to be a cost-efficient way of satisfying coverage requirements. The performance of these two formulations is also compared to a more general, non-geometry constrained third formulation proposed earlier in the literature. The formulations discussed in this paper were developed in the context of a NASA-funded project to develop a tool called Tradespace Analysis Tool for Constellations using Machine Learning (TAT-C ML), with an effort to explore a broad constellation design space during Pre-Phase A and Phase A of Earth observing missions.

While in this paper we use the term heterogeneous constellations to describe designs containing satellites at different values of altitude and inclination, other meanings of heterogeneous constellations were found in the literature: [19] and [20] respectively describe a heterogeneous constellation as an architecture comprised of different satellites of varying payload capabilities, and as a constellation that allows for inter satellite links between a remote sensing constellation and a set of small satellites dedicated for command delivery and relay data back to Earth.

The remainder of this paper is structured as follows: Section 2 contains a literature review of how evolutionary algorithms have been used to solve the constellation design problem. Section 3 presents the two new formulations (chromosome and operators) developed to explore heterogeneous designs. In this section, a brief summary of the third existing formulation is also given. Sections 4 and 5 describe the case study used to evaluate the performance of the three evolutionary formulations previously presented and explain the experimental setup to effectively compare them. Finally, Sections 6 and 7 present the results, and a discussion of the limitations of the study and opportunities for future work, respectively.

2. BACKGROUND

Evolutionary algorithms have been extensively used in constellation design for the last two decades: [2], [3] and [4] are three of the earliest works in the literature that utilize genetic algorithms to obtain a Pareto optimal set of constellation architectures, considering several conflicting objectives. These objectives are based on coverage metrics, such as percent global/regional coverage or target revisit time, and others more related to cost, such as the total number of satellites in the constellation. A few years later, Ferringer et al. [5] similarly used the NSGA-II algorithm [21] to analyze the relationship between two pairs of coverage metrics and study the sparse-coverage and resolution (temporal and spatial) trade-offs, respectively. Along the same lines, Wang et al. [6] optimized the design of a regional coverage reconnaissance satellite constellation by solving a multi-objective problem and using an evolutionary algorithm. There exist many other more recent works that also apply genetic algorithms to obtain optimal constellation designs [7], [8], [9], [10], [11], [12], [13], [14], [15].

As briefly mentioned in the introduction, most of these works restrict their design spaces to certain symmetrical constellation patterns. Some of the most common patterns are the Walker Delta [22], streets of coverage [23], the Rosette [24] and the Flower [25] configurations. Restricting the design space to just these types of constellations effectively reduces the very large space of alternatives while maintaining a high level of coverage performance. A few examples found in the literature are the following: Buzzi et al. [26] conducted an orbit and constellation design study for the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of SmallSats (TROPICS) mission by constraining the design space to Walker constellations. In similar fashion, [8] only considers symmetrical constellations with circular orbits optimizing the number of orbital planes, the number of satellites per plane, and the altitude and inclinations shared by all the satellites in the constellation. [10] co-optimizes spacecraft and orbit design for a re-configurable satellite constellation by restricting its geometry to a Walker configuration. [11] proposed a two-step optimization method to maximize coverage of satellite constellations by also assuming a Walker geometry to reduce the number of optimization variables. Similarly, [12] and [13] use the Walker constellation model to optimize, respectively, the design of a LEO satellite broadband network and a CubeSat constellation to provide continuous coverage over Europe in case of emergencies. Finally, [14] considers two types of flower constellations, 2D-LFC and 3D-LFC, which can also be described as Walker constellations, to find an optimal GNSS radio occultation constellation.

On the other hand, more general evolutionary formulations, which do not constrain the constellation geometry to symmetrical patterns, exist in the literature. [2] presents a chromosome in which the RAANs and mean anomalies of each satellite are considered separate independent variables, and the inclination is a unique variable shared by all satellites. However, they focus on circular geosynchronous orbits and, consequently, altitude and eccentricity are assumed to be constant inputs to the optimization problem. Similarly, in [3], altitude is also considered a fixed input parameter, while RAAN, mean anomaly and argument of perigee are separate independent variables for each of the satellites. Finally, eccentricity and inclination are two variables common for all satellites. In [5], the number of satellites in the constellation is fixed, and the chosen chromosome defines RAAN and mean anomaly for each satellite as a separate variable again. However, altitude is also included as a single variable shared by all satellites in the constellation, and inclination is defined as a constraint in the problem formulation. [7] breaks away from the assumption of rigidly symmetric orbits and approaches the problem from a more general design perspective searching inclinations, RAANs, and mean anomalies independently, but the semi-major axis is still kept common across the entire constellation. Finally, in [9], semi-major axis, inclination, argument of perigee, RAAN, and mean anomaly are varied as optimization design variables to search non-symmetric and circular LEO constellations with a fixed number of satellites.

All the general formulations mentioned so far that allow for different number of satellites in the final constellation [2], [3], [6], [7] use flagging in order to turn ”on” and ”off” the different participating satellites. [19] and [27] both propose

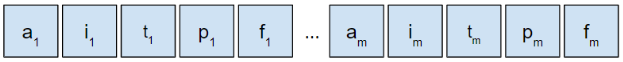

Figure 1. Fixed length chromosome encoding a hybrid architecture containing m Walker constellations.

the use of a variable-length chromosome to avoid the low-locality and redundancy of chromosomes that use Boolean flags for this purpose, since that makes the search more difficult for an evolutionary algorithm. The latter formulation –discussed in Section 3– is used in this paper because it not only uses a variable-length chromosome but also optimizes all six orbital elements (altitude, inclinations, eccentricity, argument of perigee, RAAN and mean anomaly), which are considered independent for each of the satellites in the architecture. Therefore, this evolutionary formulation is suitable for the design of heterogeneous constellations and can be considered as a benchmark, since it is the most general (least constrained) formulation possible.

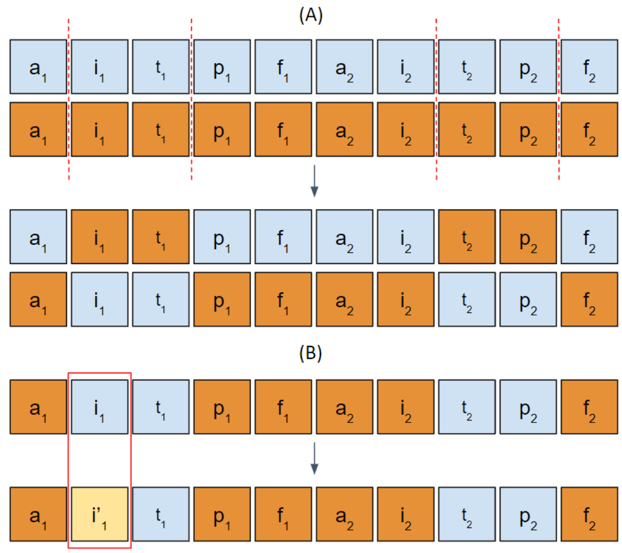

Figure 2. Operators used in formulation 1 with m two-point crossovers (A) followed by a mutation operation (B)

3. EVOLUTIONARY FORMULATIONS

The focus of this section is to describe the different evolutionary formulations considered in this paper rather than algorithmic details, since it is well known that formulations are at least as critical in driving performance as the details of the optimization algorithm used itself. Specifically, we explore two new formulations for heterogeneous constellation design, which are compared to a third formulation proposed by Hitomi and Selva [27]. For each of the formulations, we explain in detail their chromosomes and corresponding operators.

Formulation 1 - The hybrid Walker

The main idea behind this first formulation is to combine multiple Walker constellations at different altitudes and/or inclinations into a hybrid architecture. The rationale is that the resulting designs will conserve a symmetric geometry while allowing its satellites to be placed in higher/lower altitudes and more/less inclined orbits, which can be desirable for certain applications [1].

Chromosome—A fixed-length chromosome containing m 5-tuples is used to encode m Walker constellations and shown in Figure 1. Each constellation is defined by a 5-tuple (a, i, t, p, f) consisting of the altitude a, the inclination i, the number of satellites t, the number of planes p and the phasing parameter f. In this formulation, the number of satellites in each constellation can be set to 0 to effectively reduce the number of Walker constellations in the hybrid design to anywhere between 0 and m. The altitude, inclination and number of satellites variables are encoded as integer variables corresponding to the indexes of a list of allowed values. The number of planes and phasing parameters are defined as real variables taking values from 0 to 1 that are later converted to their actual values: First, the possible options for number of planes are computed depending on the number of satellites – p has to be a divisor of t. Then, the real value from 0 to 1 is mapped onto the actual number of planes option taking into account all of the possible plane options. For example, if there are $t = 3$ satellites, then the possible values for p are $p = 1, p = 3$. Therefore, any values less than 0.5 in the number of planes variable are mapped to $p = 1$, whereas values greater than 0.5 are mapped to $p = 3$.

Operators—The operators used for this first formulation are m two-point crossovers between the m walker constellations of each parent followed by a mutation operation. This procedure is illustrated in Figure 2 for two parents containing 2 Walker constellations (i.e. $m = 2$).

Formulation 2 - Hetereogeneous planes

This second formulation intends to combine multiple planes at different altitudes and inclinations without dividing the overall constellation into smaller Walker constellations. This allows us to generate designs which are less symmetric than the ones in the previous formulation, as well as to have a greater number of planes at different altitudes and inclinations. Indeed, in formulation 1, the maximum possible number of different inclination/altitude pairs would be equal to m (i.e. the maximum number of Walker constellations in the hybrid architecture).

Chromosome—Figure 3 shows the variable-length chromosome used in this second problem formulation, which encodes a heterogeneous constellation defined by the total number of satellites n, and p 2-tuples that contain the altitude and inclination of the p planes forming the constellation. Note that, unlike in Formulation 1, p is variable, and therefore the length of the chromosome depends on the architecture. All planes contain n/p satellites and they are assumed to be equally spaced in RAAN and satellites within a plane are equally spaced in mean anomaly. Similarly to what is done in Walker constellations, a phasing parameter f is defined to determine the spacing between the satellites in adjacent planes, which was assumed to be equal to 1 (i.e., the satellites in adjacent planes are separated $360°/n$). All variables in the chromosome are encoded as integer variables corresponding to the indexes of a list of allowed altitudes, inclinations and total number of satellites.

Operators—The two-point crossover operator cannot be applied to the variable-length chromosome of this second formulation because the length of the two parents can potentially be different. Instead, either a $cut\ and\ splice$ operator or a $pairing$ operator are selected with probability equal to 0.5, followed again by a mutation operation. The $cut\ and\ splice$

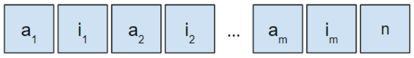

Figure 3. Variable length chromosome encoding a heterogeneous constellation defined by p planes with their corresponding inclination-altitude pair and the total number of satellites n in the constellation.

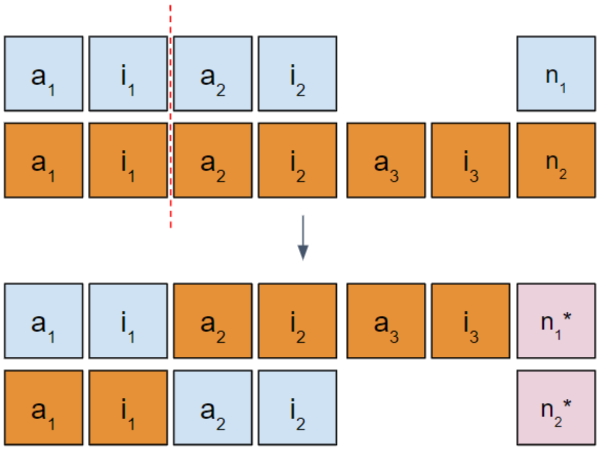

Figure 4. Cut and splice operator for the variable length chromosome from formulation 2.

operator cuts the two parent chromosomes at a randomly selected point, which is restricted to be between two planes as opposed to within a plane, and splices the cut chromosomes to create two offspring with different chromosome lengths (see Figure 4). The number of satellites for the offspring (n_i*) are chosen in the following way: if n_1 and n_2 are multiples of the resulting number of planes, choose either one randomly. If only n_1 or n_2 are multiples of the resulting number of planes, choose that value for n_i*. If neither n_1 nor n_2 are multiples of the number of planes, choose randomly any possible option from the list of allowed total number of satellites.

The *pairing* operator "pairs" each of the planes from the two parent chromosome with the fewest planes with a plane from the other parent and chooses with equal probability the values of altitude and inclination of the paired planes (see Figure 5). The *pairing* operators allows to generate combinations of altitudes/inclinations that do not yet exist within the solutions in the population, whereas the *cut and splice* operator is not able to do so.

Formulation 3 - General

Finally, we consider a much more general formulation, which provides the highest level of heterogeneity in all of the satellite's orbital elements and allows to generate constellations with no symmetries at all. With this problem formulation, we are able to explore the largest possible tradespace of constellation designs.

Chromosome— This third problem formulation uses a variable-length chromosome, shown in Figure 6, containing n 4-tuples to encode a constellation of n satellites, where each satellite is defined by a 4-tuple with its altitude, inclination, RAAN, and mean anomaly. The original formulation [27] used 6-tuples, which additionally contained the eccentricity and argument of perigee for each satellite. However, this paper focuses on assessing designs with circular orbits. All

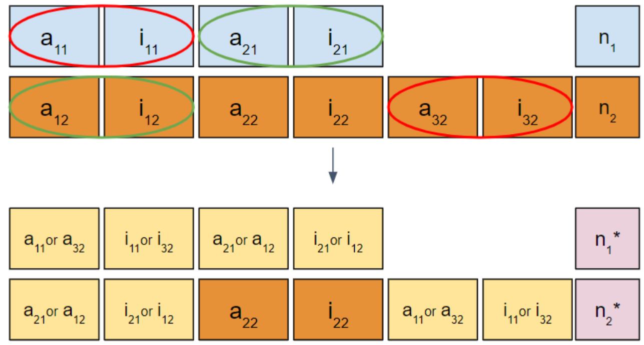

Figure 5. Pairing operator for the variable length chromosome from formulation 2.

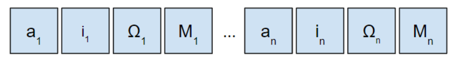

Figure 6. Variable length chromosome encoding a constellation defined by $4n$ orbital elements.

variables in the chromosome are encoded as bounded real variables.

Operators—Similar to the procedure described for the heterogeneous planes formulation, this last formulation uses either a *cut and splice* or a modified simulated binary crossover operator, selected with probability equal to 0.5, followed by a mutation operation. The modified simulated binary crossover is similar to the *pairing* operator used in formulation 2: it starts by identifying the chromosome parent containing a smaller number of satellites. Then, it pairs randomly each of the satellites of this parent with another satellite from the other parent. Finally, the paired satellites undergo a simulated binary crossover operation [28] and the unpaired satellites of the large parent retain their original values. We refer the reader to the original paper by Hitomi and Selva [27] to obtain a detailed description of the operators used in this general formulation.

4. Constellation Design Problem

The proposed formulations are applied to solve a multi-objective constellation design problem with four objectives selected in order to maximize coverage performance while minimizing overall cost. These objectives are the following:

- Minimize mean revisit time, which is the average length of the gap intervals for all the points of interest seen by the constellation.
- Minimize mean response time, which is defined as the time average of the response time. Response time corresponds to the time from when a random request is received to observe a point of interest until the constellation can actually observe it. Mean response time is known to be more sensitive to the number of planes than mean revisit time, so there can be a trade-off between these two parameters [26].
- Maximize the ratio between the number of points observed by the constellation and the total number of points in the region/s of interest. Since mean revisit time and mean response time are calculated only on observed points and do not take into account unobserved points, this third coverage metric was included as objective in order to penalize those

designs that observe fewer points in the area of interest.

- Minimize life-cycle cost, which includes launch, program, integration, ground, hardware, recurring and non-recurring costs.

Since the evolutionary formulations discussed in this paper were developed in the context of a NASA-funded project to develop a tool called Tradespace Analysis Tool for Constellations using Machine Learning (TAT-C ML), this tool was used to evaluate each of the constellation designs and obtain their respective objective values. TAT-C ML is intended to be used during early stages of the design of Earth observation missions and it is planned to be released open source within the next year. The details of both cost and orbital propagation models used for TAT-C ML are provided in [29].

The case study defined to compare the three formulations described in Section 3 has the goal to design a constellation of satellites able to monitor the Tropics and the North Pole regions, with two areas of interest located between latitudes of -30° and 30°, and 60° and 90°, respectively. This example was chosen because the disconnected region of interest is one case where heterogeneous planes might be preferable to symmetric constellations with planes at a single combination of altitude and inclination.

In this paper, only low-Earth, circular orbits are considered and each satellite in the constellation carries the same instrument with a rectangular field of view defined by a half cross-track angle of 57° and a half along-track angle of 20°. Tables 1, 2 and 3 show the possibles values/ranges of the design variables for each of the 3 formulations. In the hybrid Walker formulation, we consider the tradespace composed by up to three walker constellations, each containing up to 4 satellites, generating a hybrid architecture formed by a maximum of 12 satellites. The 3 Walkers have an altitude between 400 km and 800 km and the option to set the number of satellites equal to 0 to allow for the generation of designs that only contain 1 or 2 constellations. The first and second constellations aim to cover the area around the tropics with low inclinations, whereas the Walker containing Sun-Synchronous Orbits (SSO) is the one responsible for covering the North Pole region.

Formulation 1		
Walker	**Decision**	**Allowed values**
1	Altitudes	[400, 500, 600, 700, 800] km
	Inclinations	[0°, 10°, 20°, 30°]
	Number of Satellites	[0, 1, 2, 3, 4]
	Number of Planes	[1, 2, 3, 4]
2	Altitudes	[400, 500, 600, 700, 800] km
	Inclinations	[ISS]
	Number of Satellites	[0, 1, 2, 3, 4]
	Number of Planes	[1, 2, 3, 4]
3	Altitudes	[400, 500, 600, 700, 800] km
	Inclinations	[SSO]
	Number of Satellites	[0, 1, 2, 3, 4]
	Number of Planes	[1, 2, 3, 4]

Table 1. Possible values for the design variables in formulation 1

Similarly, the second formulation considers a heterogeneous constellation space containing up to 12 satellites distributed in up to 12 planes. Each of these planes is located at an altitude of 400km, 500km, 600km, 700km or 800km, and at an inclination of 0°, 10°, 20°, 30°, 51.6°, 90° or SSO. By breaking the Walker pattern symmetry, this formulation opens up the tradespace to a larger set of heterogeneous constellation designs.

Formulation 2	
Decision	**Allowed values**
Altitudes	[400, 500, 600, 700, 800] km
Inclinations	[0°, 10°, 20°, 30°, ISS, 90°, SSO]
Number of Satellites	[1, 2, 3, 4, 6, 8, 10, 12]
Number of Planes	[1, 2, 3, 4, 6, 8, 12]

Table 2. Possible values for the design variables in formulation 2

Finally, the design variables considered in the general formulation include the number of satellites in the constellation and each satellite's orbital elements. The number of spacecraft varies from 1 to 12, altitude from 400 to 800 km, inclination from 0° to SSO, and RAAN and mean anomaly from 0° to 360°. Since this problem formulation uses continuous variables for each of the satellite's orbital elements, the optimization algorithm explores a tradespace that contains all of the possible LEO circular constellations within the specified variables bounds.

Formulation 3		
Decision	**Lower bound**	**Upper bound**
Altitude	400 km	800 km
Inclination	0°	SSO
Number of Satellites	1	12
RAAN	0°	360°
Mean anomaly	0°	360°

Table 3. Ranges for the design variables in formulation 3

In all of the formulations, the satellites are propagated for a week using the GMAT [17] orbital propagator included in TAT-C ML. All coverage statistics are computed against a grid of 200 points spread across the two regions of interest.

5. SIMULATION SETUP

The search performed by evolutionary algorithms is stochastic, and consequently the results obtained may vary from run to run. For this reason, a total number of 30 simulations were run for each of the three formulations to be able to perform a fair statistical comparison. The ϵ-MOEA algorithm [30] was used as the baseline algorithm. This algorithm is steady-state (i.e., only one individual in the population is evolved per step) and uses an ϵ-dominance archive to maintain a well-spread set of Pareto-optimal solutions. The probability of the operators to apply crossover for formulation 1 and $cut\ and\ splice$ or $pairing$ for formulations 2 and 3 was set to $p_c = 0.9$. The probability of performing mutation for each variable (e.g., altitude, inclination) of the resulting chromosomes was set to $p_m = 0.2$ in all three evolutionary formulations proposed. An initial population of 100 random architectures was created at the beginning of each optimization run. Finally, the termination criteria was set to 1,000 function evaluations ($NFE = 1000$).

After running the total number of 90 simulations, the 90,000 architectures generated were joined together to compute the maximum and minimum values obtained for all 4 objectives, which were later used to scale between 0 and 1 the objective values of all the enumerated designs. This is common practice before proceeding to the computation of the hypervolume, an extensively used metric to evaluate the search performance of genetic algorithms [31]. The hypervolume of a set of points is the closed area (2 objectives), volume (3 objectives) or hypervolume (4+ objectives) generated by the union of all the rectangles/cuboids/hypercuboids defined by a reference vector v_{ref} (typically the anti-utopia point) and each of the points in the set. It is easy to see that only the objective vectors of non-dominated solutions contribute to the hypervolume. The higher the hypervolume indicator, the better quality has the Pareto front obtained by the evolutionary algorithm. Hypervolume was computed using the Pagmo Python library [32] after each function evaluation for every single run using the reference point $v_{ref} = [1, 1, 1, 1]$, which corresponds to the maximum (worst) possible value for each scaled objective.

In the following Section, the hypervolume indicator is used to assess the differences in performance of the 3 evolutionary formulations proposed in Section 3. Particularly, the non-parametric Wilcoxon rank sum test with a significance level of 0.05 was used to compare the differences in hypervolume between all 3 formulations. Additionally, the compositions of the 3 resulting final Pareto fronts are analyzed and compared against each other.

6. RESULTS

Figure 7 shows the evolution of hypervolume with respect the number of function evaluations (NFE) for each of the formulations proposed in Section 3. The thick solid line represents the average hypervolume of the 30 runs and the lighter shaded area shows one standard deviation from the mean value. It is observed that the third formulation starts performing better than the other two, providing a statistically greater hypervolume ($p < 0.0444$ and $p < 0.0428$) than formulations 1 and 2 between 7 and 76 and 1 and 87 NFE, respectively. However, this first portion of the plot simply corresponds to the first 100 designs randomly generated. Then, between 100 and 200 NFE, the hybrid Walker becomes the best formulation and continues to outperform the other two until the end of the run. Conversely, the heterogeneous planes formulation starts performing worse than the other two but, around 300 NFE, it becomes better than the most general formulation in mean value. The variance in hypervolume for this second formulation is larger than the other two and, despite its mean hypervolume is higher than the general formulation at 1000 NFE, a final lower hypervolume is obtained in some of the runs. The results from the Wilcoxon rank sum test showed that formulation 1 maintains a statistically greater hypervolume ($p < 0.0476$ and $p < 0.0444$) than formulations 2 and 3 after 138 and 140 function evaluations, respectively. Similarly, formulation 2 offers a statistically greater hypervolume ($p < 0.04926$) than formulation 3 from 433 function evaluations on.

Additionally, Figure 8 plots the ratio between the number of runs that reach the values of 0.85 and 0.87 hypervolume

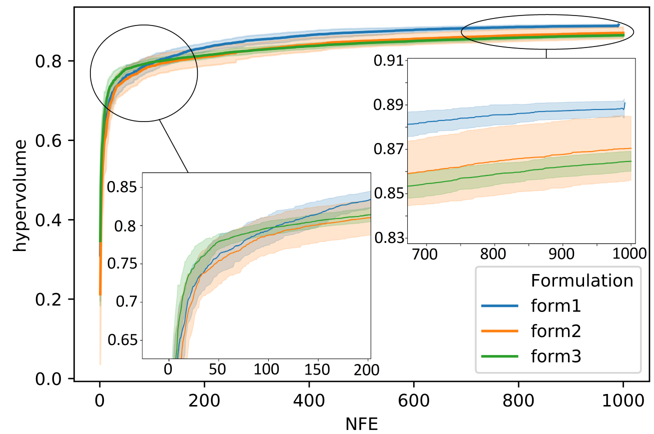

Figure 7. Hypervolume as a function of the number of function evaluations for each of the three formulations.

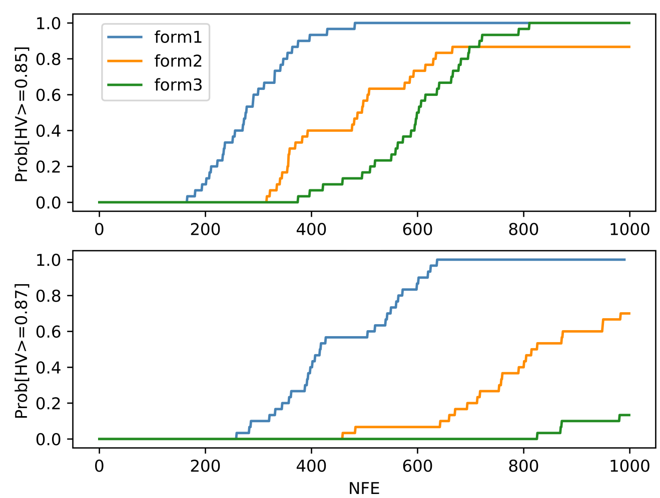

Figure 8. Probability of attaining 0.85 and 0.87 HV for each of the three formulations.

and the total number of runs (i.e., 30) with respect to the number of function evaluations. In other words, this figure shows the probability of attaining such hypervolume values by different values of NFE for each of the three evolutionary formulations. In the bottom plot, 0.87 corresponds to the maximum value of hypervolume achieved by all three evolutionary formulations. It can be observed that the hybrid Walker formulation performs the best, with a probability of reaching 0.87 hypervolume of almost 60% within 400 NFE and 100% within less than 700 NFE. On the other hand, the heterogeneous planes formulation only reaches 0.87 hypervolume in roughly 70% of the runs, while the most general formulation only achieves that threshold value with a probability lower than 20%. Conversely, in the top plot of Figure 8 it is observed that formulations 2 and 3 attain 0.85 hypervolume with 80% and 100% probability, respectively. Even though the heterogeneous planes formulation provides higher probability of attaining 0.85 hypervolume than the general formulation for low NFEs, at 700 NFE formulation 3 becomes more reliable, reaching a value of 100% probability within only 800 NFE. This is due to the high variance of formulation 2 observed in Figure 7, where it is shown that all the runs for formulation 3 reach a hypervolume greater than 0.85, but some of the runs in formulation 2 do not.

The behaviour seen in Figures 7 and 8 can be explained by two main reasons:

- The selected constellation design problem, despite considering two different regions of interest at different latitudes (Tropics and North Pole), can still benefit from architecture designs mixing symmetric constellations at different inclinations (i.e., hybrid Walkers), since the points of interest are distributed uniformly across longitude.
- The three formulations considered have different design spaces. As mentioned in Section 3, the tradespace of the third formulation contains all possible circular LEO constellations between 400km and 800km of altitude. Therefore, this formulation reasonably needs more time to explore a larger tradespace and converge to the symmetric designs offered by the hybrid Walker formulation, which logically needs a smaller number of function evaluations to do so. Finally, the heterogeneous planes formulation lies in between formulations 1 and 3 regarding design space size and symmetry of the constellations generated. Consequently, it offers higher search performance than the general formulation in the long run despite having smaller values of hypervolume after the random initialization at the beginning of the genetic algorithm. In summary, having a larger design space is a double-edged sword: on the one hand, it can help finding more novel constellation designs one could not think of. On the other hand, it will take more computation time to find constellation designs that one already knows are good, such as a symmetric Walker constellation.

In an effort to compare the final non-dominated designs or Pareto front of the three studied formulations, Figure 9 shows the projection of the three 4-dimensional Pareto fronts onto the six different planes generated by all possible pairs of the four objectives considered in the optimization problem. Even though these six plots do not show actual 2-dimensional Pareto fronts but a projection of three 4-dimensional Pareto fronts, it can be noted that for most of the 2-D scatter plots, formulations 1 and 2 outperform formulation 3 as they both obtain better projected ”Pareto” fronts.

While Figure 9 plots the Pareto fronts on the objective space, Table 4 aims to describe the obtained non-dominated solutions on the design space. In 3 representative runs (one for each formulation), the Pareto fronts obtained contain 65, 58 and 50 constellation designs for formulation 1, 2 and 3 respectively. In Table 4, just a small illustrative sample of this set of architectures is shown.

The hybrid Walker formulation effectively generates architectures that mix Walkers at low inclinations to monitor the Tropics and SSO Walkers to track the North Pole. These hybrid Walker constellations grant 100% coverage but also single Walker designs are found by the genetic algorithm, which are cheaper but provide worse coverage statistics.

Similarly, the heterogeneous planes formulation enumerates designs that mix planes at lower inclinations (20° and 30°) with polar and SSO planes to successfully observe both the tropical and polar regions. More affordable options with just one SSO plane are also found in the Pareto front, which provide 100% coverage at a much lower cost. Finally, constellations with just one plane at very low inclinations that have very good values of mean response and revisit times are also enumerated. However, these constellations only cover a bit over 30% of the whole area of interest.

Finally, the third and most general formulation enumerates designs that mix satellites at low inclinations (roughly between 15° and 30°) and high inclinations (roughly between 70° and 90°) to again track both areas of interest in our problem. As in formulation 1, cheap solutions with just 1 satellite at a very low inclination are found in the final Pareto front. Almost all the non-dominated designs in all of the formulations contain high values of altitude between 700km and 800km. This happens because the cost model used to evaluate each of the designs is not sensitive to altitude variations. Consequently, higher altitudes dominate lower altitudes, as they provide better coverage statistics “for free”.

7. CONCLUSIONS

This work introduced two new evolutionary formulations (hybrid Walker and heterogeneous planes) to study heterogeneous constellations mixing satellites at different altitudes and inclinations. So far, this part of the constellation design space had been understudied and this paper showed that it has potential to be a cost-efficient way of satisfying mission requirements for certain Earth observing applications. By solving a constellation design problem optimizing mean revisit time, mean response time, coverage and cost, it was shown that these two formulations obtain a higher convergence rate and diversity (measured by the final hypervolume) than a third formulation already existing in the literature, which used a variable-length chromosome with continuous variables representing the 4 orbital elements of each satellite. Additionally, it was observed that the hybrid Walker formulation performs better than the heterogeneous planes one. However, this behaviour could be related to the fact that the constellation design problem considered, despite having two regions of interest, it is still fairly symmetric and therefore it can still benefit from the symmetries offered by the hybrid Walker configuration. An opportunity for future work would be to study a wide range of constellation design problems to study how much the performance of the proposed formulations is problem dependent.

The termination criteria of the multi objective evolutionary algorithm was set to a maximum number of function evaluations equal to 1,000. Considering tradespaces that have millions or even billions of possible architectures, 1,000 function evaluations are clearly not enough for the algorithm to converge. This can be seen in Figure 7, where the hypervolume keeps increasing until the end of the simulation for all three formulations. Moreover, the three studied formulations explore tradespaces of different sizes. More specifically, the design space in formulation 3 is larger than the one in formulation 2, whose tradespace size is larger than the one in formulation 1. Consequently, the general formulation might require more function evaluations to converge than any of the other two, and so does formulation 2 with respect to 1. For all these reasons, another opportunity for future work would be analyzing the same problem for a larger number of function evaluations to see if the formulations exploring larger tradespaces eventually find better designs.

There is also room to improve the chromosome representation of both hybrid Walker and heterogeneous planes formulations. On the one hand, the fixed-length chromosome described for the hybrid Walker approach could be made variable-length to avoid having to set the number of satellites t to 0 to turn ”off” a certain Walker, which constitutes a source of redundancy in the chromosome representation. Moreover, this fix would allow to not having to specify the maximum number of Walker constellations in the hybrid architecture

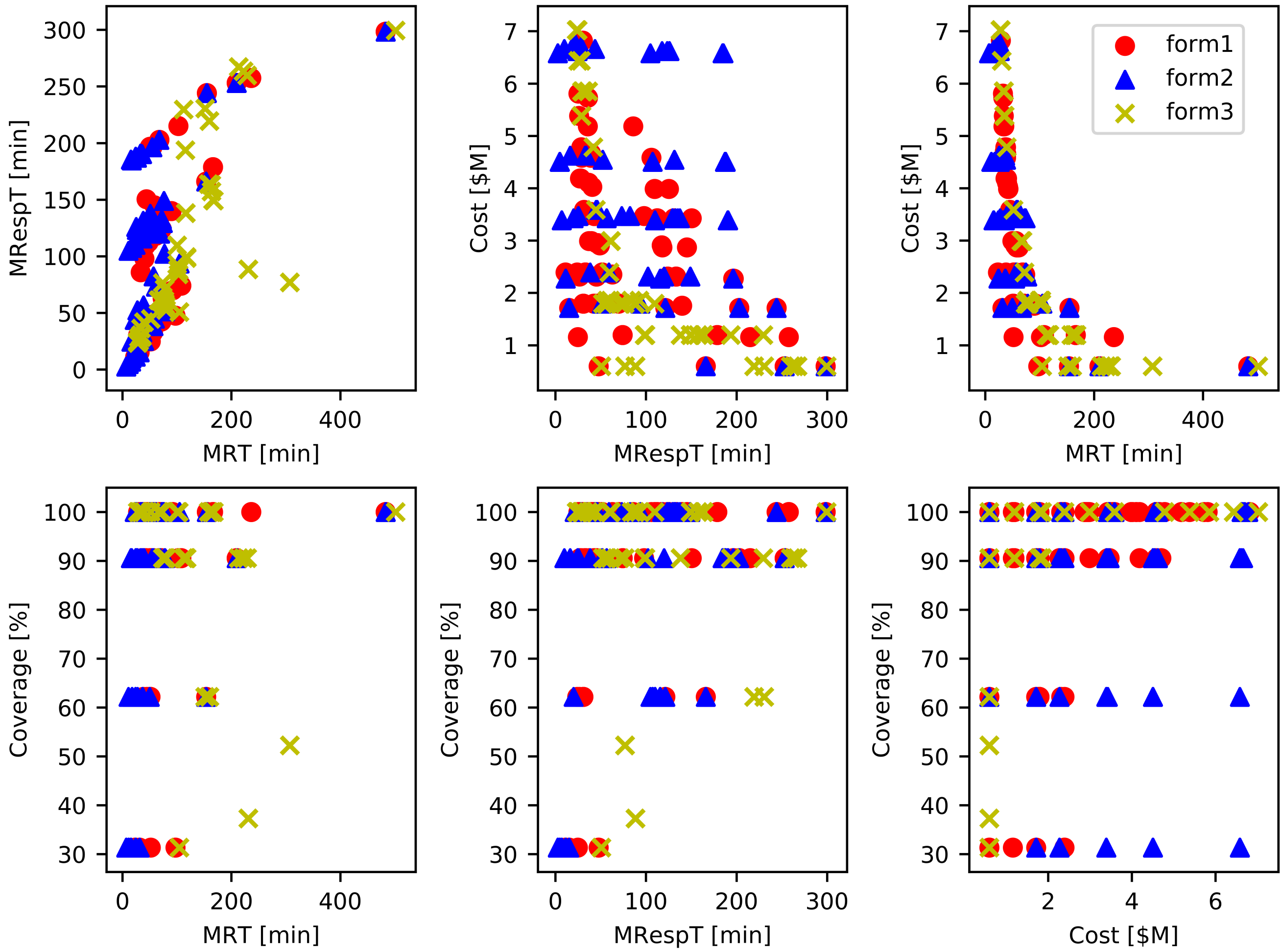

Figure 9. Comparison of Pareto fronts obtained in a representative run for each of the three formulations.

Decisions	**Objectives** (MRT, MRespT, Cost, Cov)
Formulation 1 - Hybrid Walker	
(800km 20° t:4 p:4 f:0) (800km SSO t:3 p:1 f:0) 7 sats	(2408.19s, 2200.86s, $4.11M, 100%)
(400km 0° t:1 p:1 f:0) 1 sat	(5844.69s, 2857.73s, $0.6M, 31.34%)
(800km 20° t:4 p:2 f:0) (800km 51.6° t:4 p:1 f:0) 8 sats	(2112.14s, 1748.14s, $4.58M, 90.55%)
(800km 20° t:4 p:4 f:0) (600km 51.6° t:2 p:1 f:0) (800km SSO t:4 p:4 f:3) 10 sats	(1957.52s, 1519.79s, $5.81M, 100%)
(800km 20° t:4 p:2 f:0) (800km 51.6° t:4 p:4 f:3) (800km SSO t:4 p:1 f:0) 12 sats	(1728.78s, 1821.48s, $6.82M, 100%)
Formulation 2 - Heterogeneous planes	
(800km SSO) (800km 30°) 4 sats	(4576.04s, 8932.34s, $2.31M, 100%)
(800km SSO) 1 sat	(28984.02s, 17902.67s, $0.6M, 100%)
(800km SSO) (800km 30°) 12 sats	(1504.54s, 7547.62s, $6.62M, 100%)
(800km 30°) (700km 90°) (700km 20°) (800km 20°) 12 sats	(1513.80s, 1701.70s, $6.7M, 100%)
(400km 0°) 8 sats	(666.39s, 299.76s, $4.5M, 31.34%)
Formulation 3 - General	
(790.83km 20.19°) (759.193km 79.75°) (748.54km 31.13°) (717.03km 26.77°) (795.53km 98.19°) (790.64km 5.34°) (745.1km 27.79°) (716.79km 26.14°) (776.59km 26.18°) 9 sats	(2079.68s, 1728.04s, $5.38M, 100%)
(732.10 20.19°) (785.13 14°) (784.15 23.43°) 3 sats	(4633.70s, 3644.62s, $1.85M, 90.55%)
(734.51 2.28°) 1 sat	(6278.17s, 3048.05s, $0.6M, 31.34%)
(717.04km 26.77°) (777.33km 15.99°) (737.61km 26.57°) (799.05km 20.26°) (696.33km 21.71°) (776.21km 79.63°) (702.03km 85.87°) (623.84km 25.07°) (738.89km 72.72°) (727.69km 82.93°) (773.5km 12°) (698.31km 90.54°) 12 sats	(1673.67s, 1476.89s, $7.02M, 100%)

Table 4. Sample of architectures in the Pareto front obtained in a randomly selected run for each of the three formulations

(e.g. in the constellation design problem presented in Section 4, the maximum number of Walkers was set to 3). On the other hand, in the heterogeneous planes formulation, the phasing parameter f was assumed to be equal to 1. Adding an extra parameter to the chromosome to characterize the phasing between adjacent planes besides the number of satellites and the altitude/inclinations pairs would open a bit more the design space, which could lead to find better unexplored solutions for the current evolutionary formulation implementation. Furthermore, our heterogeneous plane approach assumes equal number of satellites in each plane. This design constraint could be also removed to include even less symmetric architectures. Lastly, to further open up the tradespace and to better exploit the advantages of constellations that mix satellites at different altitudes and inclinations, future work may also include the possibility of considering constellations composed by spacecraft carrying distinct instruments.

Finally, the creation of other operators and the use of credit assignment strategies in multi objective adaptive operator selection [33], [34], [35] could help get better convergence rates but, again, the algorithmic details were not the focus of this paper.

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REFERENCES

[1] H. W. Lee, S. Shimizu, S. Yoshikawa, and K. Ho, "Satellite constellation design methodology for optimal regional coverage," *arXiv preprint arXiv:1910.00672*, 2019.

[2] W. Mason, V. Coverstone-Carroll, and J. Hartmann, "Optimal earth orbiting satellite constellations via a pareto genetic algorithm," in *AIAA/AAS Astrodynamics Specialist Conference and Exhibit*, 1998, p. 4381.

[3] G. Confessore, M. Di Gennaro, and S. Ricciardelli, "A genetic algorithm to design satellite constellations for regional coverage," in *Operations Research Proceedings*. Springer, 2001, pp. 35–41.

[4] M. Asvial, R. Tafazolli, and B. G. Evans, "Non-GEO satellite constellation design with satellite diversity using genetic algorithm," *20th AIAA International Communication Satellite Systems Conference and Exhibit*, no. May 2002, pp. 1–7, 2002.

[5] M. P. Ferringer and D. B. Spencer, "Satellite constellation design tradeoffs using multiple-objective evolutionary computation," *Journal of Spacecraft and Rockets*, vol. 43, no. 6, pp. 1404–1411, 2006.

[6] L. Wang, Y. Wang, K. Chen, and H. Zhang, "Optimization of regional coverage reconnaissance satellite constellation by nsga-ii algorithm," in *2008 International Conference on Information and Automation*. IEEE, 2008, pp. 1111–1116.

[7] W. R. Whittecar and M. P. Ferringer, "Global coverage constellation design exploration using evolutionary algorithms," *AIAA/AAS Astrodynamics Specialist Conference 2014*, no. August, pp. 1–20, 2014.

[8] I. Meziane-Tani, G. Métris, G. Lion, A. Deschamps, F. T. Bendimerad, and M. Bekhti, "Optimization of small satellite constellation design for continuous mutual regional coverage with multi-objective genetic algorithm," *International Journal of Computational Intelligence Systems*, vol. 9, no. 4, pp. 627–637, 2016.

[9] T. Savitri, Y. Kim, S. Jo, and H. Bang, "Satellite Constellation Orbit Design Optimization with Combined Genetic Algorithm and Semianalytical Approach," *International Journal of Aerospace Engineering*, vol. 2017, no. May, 2017.

[10] S. W. Paek, S. Kim, and O. de Weck, "Optimization of reconfigurable satellite constellations using simulated annealing and genetic algorithm," *Sensors (Switzerland)*, vol. 19, no. 4, 2019.

[11] C. Han, S. Bai, S. Zhang, X. Wang, and X. Wang, "Visibility optimization of satellite constellations using a hybrid method," *Acta Astronautica*, no. January, pp. 0–1, 2019. [Online]. Available: https://doi.org/10.1016/j.actaastro.2019.01.025

[12] D. Yan, P. You, C. Liu, S. Yong, and D. Guan, "Constellation multi-objective optimization design based on QoS and network stability in LEO satellite broadband networks," *KSII Transactions on Internet and Information Systems*, vol. 13, no. 3, pp. 1260–1283, 2019.

[13] L. Agresti, S. Corpino, P. F. Stesina, P. D. B. Tridon, P. L. Savioli, and A. D'Ottavio, "Multi objective genetic algorithm optimization of a cubesats constellation to ensure communication services during natural disasters over europe," Ph.D. dissertation, Politecnico di Torino, 2019.

[14] X. Xu, Y. Han, J. Luo, J. Wickert, and M. Asgarimehr, "Seeking optimal GNSS radio occultation constellations using evolutionary algorithms," *Remote Sensing*, vol. 11, no. 5, pp. 1–17, 2019.

[15] F. Keyvani and S. H. Torabi, "Design and simulation of regional navigation constellation with optimized mean DOP based on hybrid GEO and IGSO satellites," *International Journal of Aviation, Aeronautics, and Aerospace*, vol. 6, no. 2, 2019.

[16] A. G. Inc. (2019) Systems tool kit (stk). [Online]. Available: https://www.agi.com/products

[17] S. P. Hughes, R. H. Qureshi, S. D. Cooley, and J. J. Parker, "Verification and validation of the general mission analysis tool (gmat)," in *AIAA/AAS astrodynamics specialist conference*, 2014, p. 4151.

[18] L. Maisonobe, V. Pommier, and P. Parraud, "Orekit: An open source library for operational flight dynamics applications," in *4th International Conference on Astrodynamics Tools and Techniques*, 2010, pp. 3–6.

[19] K. Mott and J. Black, "Model-Based Heterogeneous Optimal Space Constellation Design," *2018 21st International Conference on Information Fusion, FUSION 2018*, pp. 602–609, 2018.

[20] I. Sanad and D. G. Michelson, "A Framework for Heterogeneous Satellite Constellation Design for Rapid Response Earth Observations," *IEEE Aerospace Conference Proceedings*, vol. 2019-March, pp. 1–10, 2019.

[21] K. Deb, A. Member, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multi-objective genetic algorithm:NSGAII," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, 2002.

[22] J. G. Walker, "Circular orbit patterns providing continuous whole earth coverage," ROYAL AIRCRAFT ESTABLISHMENT FARNBOROUGH (UNITED KINGDOM), Tech. Rep., 1970.

[23] W. S. Adams and L. Rider, "Circular polar constellations providing continuous single or multiple coverage above a specified latitude," *Journal of the Astronautical Sciences*, vol. 35, pp. 155–192, Jun. 1987.

[24] A. H. Ballard, "Rosette constellations of earth satellites," *IEEE Transactions on Aerospace and Electronic Systems*, no. 5, pp. 656–673, 1980.

[25] D. Mortari, M. P. Wilkins, and C. Bruccoleri, "The flower constellations," *Journal of Astronautical Sciences*, vol. 52, no. 1, pp. 107–127, 2004.

[26] P. G. Buzzi, D. Selva, N. Hitomi, and W. J. Blackwell, "Assessment of constellation designs for earth observation: Application to the TROPICS mission," *Acta Astronautica*, vol. 161, pp. 166–182, aug 2019.

[27] N. Hitomi and D. Selva, "Constellation optimization using an evolutionary algorithm with a variable-length chromosome," *IEEE Aerospace Conference Proceedings*, vol. 2018-March, pp. 1–12, 2018.

[28] K. Deb and R. B. Agrawal, "Simulated binary crossover for continuous search space," *Complex systems*, vol. 9, no. 2, pp. 115–148, 1995.

[29] J. Le Moigne, P. Dabney, O. De Weck, V. Foreman, P. Grogan, M. Holland, S. Hughes, and S. Nag, "Tradespace analysis tool for designing constellations (TAT-C)," *International Geoscience and Remote Sensing Symposium (IGARSS)*, vol. 2017-July, pp. 1181–1184, 2017.

[30] M. Laumanns, L. Thiele, K. Deb, and E. Zitzler, "Combining convergence and diversity in evolutionary multiobjective optimization," *Evolutionary computation*, vol. 10, no. 3, pp. 263–282, 2002.

[31] C. A. C. Coello, G. B. Lamont, and D. A. V. Veldhuizen, *Evolutionary Algorithms for Solving Multi-Objective Problems*. Springer US, 2007.

[32] F. Biscani, D. Izzo, W. Jakob, GiacomoAcciarini, M. Märtens, M. C, A. Mereta, C. Kaldemeyer, S. Lyskov, S. Corlay, G. Acciarini, B. Pritchard, K. Manani, Acxz, J. Mabille, A. Huebl, M. López-Ibáñez, Jakirkham, J. Lee, Hulucc, Polygon, L. Č. Zajc, J. Adler, J. Travers, J. Jordan, I. Smirnov, H. Nguyen, F. Lema, E. O'Leary, and A. Mambrini, "esa/pagmo2: pagmo 2.11.4," sep 2019. [Online]. Available: https://doi.org/10.5281/zenodo.3464510

[33] N. Hitomi, H. Bang, and D. Selva, "Adaptive Knowledge-Driven Optimization for Architecting a Distributed Satellite System," *Journal of Aerospace Information Systems*, vol. 15, no. 8, pp. 485–500, jun 2018. [Online]. Available: https://doi.org/10.2514/1.I010595

[34] N. Hitomi and D. Selva, "Incorporating expert knowledge into evolutionary algorithms with operators and constraints to design satellite systems," *Applied Soft Computing Journal*, vol. 66, no. May 2018, pp. 330–345, 2018.

[35] ——, "A classification and comparison of credit assignment strategies in multiobjective adaptive operator selection," *IEEE Transactions on Evolutionary Computation*, vol. 21, no. 2, pp. 294–314, 2017.

BIOGRAPHY

Pau Garcia Buzzi *is a PhD student in Aerospace Engineering at the College of Engineering at Texas A&M University. Prior to Texas A&M, Pau received a double Bachelor's degree in industrial and telecommunications engineering from Universitat Politecnica de Catalunya in Barcelona, Spain. In August 2018, he graduated with a Master's of Science degree in Aerospace Engineering at the Sibley School of Mechanical and Aerospace Engineering from Cornell University. He is particularly interested in the coverage analysis and constellation design for Earth observation missions. During his Master's degree, Pau worked under the supervision of Dr. Daniel Selva as part of the orbit and constellation design team of the NASA-funded Time-Resolved Observations of Precipitation Structure and Storm Intensity (TROPICS) mission. This experience provided him with valuable experience and insight into the orbit analysis and design of large constellations of Earth observing satellites. His main research interest is applying machine learning and optimization techniques to the orbit and constellation design of future space missions.*

Daniel Selva *is an Assistant Professor in the department of Aerospace Engineering at Texas A&M University, where he directs the Systems Engineering, Architecture, and Knowledge (SEAK) Lab. His research interests focus on the application of knowledge engineering, global optimization and machine learning techniques to systems engineering and architecture, with a strong focus on space systems. Before doing his PhD in Space Systems at MIT, Daniel worked for four years in Kourou (French Guiana) as an avionics specialist within the Ariane 5 Launch team. Daniel has a dual background in electrical engineering and aerospace engineering, with degrees from MIT, Universitat Politecnica de Catalunya in Barcelona, Spain, and Supaero in Toulouse, France. He is a member of the European Space Agency's Advisory Committee for Earth Observation and of the AIAA Intelligent Systems Technical Committee.*